IKRAMUL HASAN MORAL

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Final-year CSE student (CGPA 3.94) who builds production web platforms and does applied machine learning research. Sole remote engineer for a live e-learning platform, teaching assistant for four lab courses, two-time university champion.

EDUCATION

B.Sc. in Computer Science and Engineering | *United International University (UIU), Dhaka* — Expected Jan/Feb 2027

- CGPA 3.94/4.0; 8 merit scholarships: six at 100% (top 3% each trimester), one at 50%, one at 25%
- Thesis: Lorenz-1960 PINN, solving the ODE system from the equations alone; supervisor: Dr. Muhammad Nomani Kabir
- Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, DBMS, Software Engineering, Machine Learning, AI, HCI, Mobile Application Development, System Analysis & Design

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, C++, Java, SQL, HTML/CSS
Frontend: React 19, Next.js 15, React Native, Expo, Tailwind CSS, Radix UI, Zustand, TanStack Query, Redux
Backend: Node.js, Express, NestJS, MongoDB, PostgreSQL, Redis, Supabase, Firebase, REST, OpenAPI/Swagger, Zod, JWT, Socket.io, WebRTC
ML & Data: PyTorch, scikit-learn, XGBoost, SMOTE/imbalanced-learn, pandas, NumPy, SciPy, Matplotlib
Tools: Git, GitHub Actions, Docker, Vercel, Railway, Linux, Postman, Stripe, Cloudinary, Arduino

EXPERIENCE

Undergraduate Teaching Assistant | *United International University, Dept. of CSE* — Aug 2025 - Present

- Support DBMS, DSA-I, ICS, and SWE lab courses; weekly lab sessions and tutorials on graphs, DP, SQL, normalization
- Design and grade programming assignments; hold weekly office hours and code reviews

Software Engineer (Sole Engineer, Remote) | *RimonsIELTS, IELTS e-learning platform* — 2025 - Present

- Build and operate 4 live apps solo: Next.js 15 student portal, Express + TypeScript REST API, React 19 admin dashboard, Python central service
- Integrated Vimeo/Zoom video streaming, AI study features, cloud file storage, and course/exam management
- Ship with Zod validation on every boundary, OpenAPI docs, JWT auth, server tests, and go-live checklists
- Long-term collaboration: rebuilt their first MERN site (2022) into today's production platform

RESEARCH

Class Imbalance Study on UNSW-NB15 | *university term paper* — 2025 - 2026

- Experiment grid: binary + multiclass tasks; LR, RF, XGBoost; no balancing vs class weighting vs SMOTE
- Seeded, leakage-proof pipeline; macro-F1, ROC-AUC, and G-Mean with per-class analysis

Undergraduate Thesis: Lorenz ODE System via Optimal ANN Architectures | *team lead of 5, FYDP-II defended, manuscript submitted to ICEEICT 2027* | code — 2025 - 2026

- Residual-only PINN (11,283 params) with a hard-constrained initial condition; within 3e-05 of a DOP853 reference, trained in under six CPU minutes
- Nine-architecture sweep plus a windowed causal-weighting study for longer horizons; 32 trained models archived with SHA-256 verification and 41 tests
- Loss falls 7.5 orders of magnitude to 6.8e-09; residual generalisation gap 1.000-1.009 across all runs

NB Instance Filtering and Tree-Based Attribute Weighting | *manuscript written, not yet submitted* | code — 2026

- Compares NB instance filtering and decision-tree attribute weighting, separately and in both orders
- All supervised steps refit inside each training fold; scored on held-out rows across ten seeds (Weka J48)

Physics-Residual Detection of CAN Masquerade Attacks | *physics-informed vs data-driven, ROAD dataset* | code — 2026

- Tests whether physics-residual signals attribute masquerade attacks on in-vehicle CAN traffic better than purely data-driven baselines

ANN Modeling of Hybrid Nanofluid Boundary Layer Flow | *independent scientific ML study* — 2026

- 9-layer network trained with Levenberg-Marquardt optimization on ~32,400 physics-generated samples
- Validated against numerical solutions of the flow equations using MSE, RMSE, and R^2 metrics

PROJECTS

Advyon - Legal-Tech Platform | *React, Express, TypeScript, MongoDB* | github | live — 2025

- Led 5-member team through full SDLC; Champion at UIU CSE Project Show (Software Engineering Lab)
- Case management, document workflows, community forum, smart intake, AI-assisted legal insights
- Express + TypeScript service layer with JWT auth, role-based access, and legal database services

BriefLi - AI Shortlink Manager | *React, TypeScript, Vite, Supabase, Vercel, Gemini API* | github | live — 2025 - Present

- Live shortlink dashboard: AI slug suggestions, click tracking, public + authenticated link spaces, trash/restore
- Vercel serverless API; Supabase Postgres with SQL migrations and unique-slug constraints

Fall Detection Wearable | *Arduino/C++, Python ML, Node.js, JavaScript* | github — 2025

- 4-layer IoT system: wearable firmware, Python ML fall classifier, Node.js alert server, caregiver web dashboard
- Real-time fall classification from accelerometer streams with instant alert delivery

Other Projects

- Shera Bot (AI agent runtime, TypeScript monorepo) | VoiceKey (offline Whisper voice typing, Linux/Windows) | Paradox Prism (anonymous reporting + steganography) | PrintSlot (campus print-shop app)
- Carried (no-account book-line cards) | ODE 3D Orbit (interactive 3D ODE explorer) | Skill Paradox (skill-exchange platform) | Meme Research Console (live study instrument) | Email Templates (live tool)

ACHIEVEMENTS & LEADERSHIP

- Champion & Team Lead, SE Lab, UIU CSE Project Show (Advyon); Champion, CSE Week Programming Contest 2024 (80+ participants)
- 500+ problems solved on Codeforces, LeetCode, and HackerRank (2022-2025; no longer active in contests)
- 1st Runner-Up, Advanced OOP Project Showcase 2023; 6th Runner-Up, System Analysis & Design Lab Showcase 2024
- Peer mentor, UIU CSE community; exam study packs used by classmates across the department